CBIT/IT-1/EDAV/2026/CEP-28

Analyze forest cover trends and distribution over Indian states

Roll No: 160125737012 & 160125737019
Subject: EDAV - Exploratory Data Analysis and Visualization
Dataset: forest_cover_india_states_2017_2023_final.csv (Official ISFR)

Problem Statement

Analyze forest cover trends and distribution over Indian states using official Forest Survey of India data.

Official Questions (as per PS-28 sheet):

- Q1: Calculate mean forest cover area (BL-3)
- Q2: Filter states by forest cover type (BL-3)
- Q3: Fill missing forest cover data (BL-4)
- Q4: Group data by region and cover type (BL-4)
- Q5: Visualize forest cover change using plots (BL-5)

Forest Cover Types (Theory):

- VDF - Very Dense Forest (Canopy >70%)
- MDF - Moderately Dense Forest (Canopy 40-70%)
- OF - Open Forest (Canopy 10-40%)
- Source: ISFR Classification by FSI

Dataset Link (Official Source - Verified)

- **Source:** Forest Survey of India (FSI), MoEFCC, Govt of India
- **Report:** India State of Forest Report (ISFR) 2017-2023
- **Official Link:** https://fsi.nic.in/forest-report-2023
- **Official PDF:** https://fsi.nic.in/uploads/isfr2023/isfr_book_eng-vol-2_2023.pdf
- **File Used:** forest_cover_india_states_2017_2023_final.csv (36 States/UTs, 17 columns)
- **Library Used:** pandas + matplotlib (as per Guideline 6)

Objective

1. To calculate average forest cover area across states (Q1)
2. To filter states based on VDF/MDF/OF cover type (Q2)
3. To handle and fill missing values in forest data (Q3)

4. To group data by Region and dominant cover type for analysis (Q4)
5. To visualize forest cover trends and changes from 2017-2023 (Q5)

In [14]:
```
# Q1: Calculate mean forest cover area (BL-3)
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('forest_cover_india_states_2017_2023_final.csv')

mean_2023 = df['Forest_Cover_2023_km2'].mean()
mean_2017 = df['Forest_Cover_2017_km2'].mean()
print(f"Mean Forest Cover 2023: {mean_2023:.2f} km2")
print(f"Mean Forest Cover 2017: {mean_2017:.2f} km2")
```

```
Mean Forest Cover 2023: 19870.63 km2
Mean Forest Cover 2017: 20235.79 km2
```

In [15]:
```
# Q2: Filter states by forest cover type (BL-3)
# VDF > 5000
print("States with VDF > 5000 km2:")
print(df[df['VDF_2023_km2'] > 5000][['State_UT','VDF_2023_km2']])

# OF dominant states
print("\nStates where OF is dominant:")
print(df[df['OF_2023_km2'] > df['MDF_2023_km2']][['State_UT','OF_2023_km2']].hea
```

```
States with VDF > 5000 km2:
                    State_UT  VDF_2023_km2
1           Arunachal Pradesh      20985.32
4                Chhattisgarh       7416.57
13             Madhya Pradesh       7021.31
14                Maharashtra       9865.62
19                     Odisha       7224.42
27                Uttarakhand       5041.18
29  Andaman & Nicobar Islands       5682.80

States where OF is dominant:
         State_UT  OF_2023_km2
0  Andhra Pradesh     14363.50
2           Assam     15359.30
3           Bihar      3861.24
5           Delhi       135.26
6             Goa      1120.83
```

Interpretation (BL-3 Apply): 7 states have Very Dense Forest >5000 km2, Arunachal highest (20985 km2) due to Northeast Himalayan region. OF dominant in 15+ states showing open forest is major type in plains. Filter function `df[condition]` used successfully.

In [16]:
```
# Q3: Fill missing forest cover data (BL-4)
print("Before:\n", df.isnull().sum().head(10))

df['Geographical_Area_km2'] = df['Geographical_Area_km2'].fillna(df['Geographica
df['Forest_Cover_2017_km2'] = df['Forest_Cover_2017_km2'].fillna(df['Forest_Cove
df['Forest_Cover_Change_2017_2019_km2'] = df['Forest_Cover_Change_2017_2019_km2'
df['Forest_Cover_Change_2017_2023_km2'] = df['Forest_Cover_Change_2017_2023_km2'
df['Forest_Cover_2023_pct'] = df['Forest_Cover_2023_pct'].fillna(df['Forest_Cove
df['Data_Note'] = df['Data_Note'].fillna('No Note')
```

```
print("Data_Note column after filling:\n", df['Data_Note'].head(10))
print("\nAfter:\n", df.isnull().sum().head(10))
```

```
Before:
 State_UT                             0
Region                                0
Geographical_Area_km2                 1
Forest_Cover_2017_km2                 1
Forest_Cover_2019_km2                 0
Forest_Cover_2021_km2                 0
Forest_Cover_2023_km2                 0
Forest_Cover_Change_2017_2019_km2     1
Forest_Cover_Change_2019_2021_km2     0
Forest_Cover_Change_2021_2023_km2     0
dtype: int64
Data_Note column after filling:
 0    No Note
1    No Note
2    No Note
3    No Note
4    No Note
5    No Note
6    No Note
7    No Note
8    No Note
9    No Note
Name: Data_Note, dtype: str

After:
 State_UT                             0
Region                                0
Geographical_Area_km2                 0
Forest_Cover_2017_km2                 0
Forest_Cover_2019_km2                 0
Forest_Cover_2021_km2                 0
Forest_Cover_2023_km2                 0
Forest_Cover_Change_2017_2019_km2     0
Forest_Cover_Change_2019_2021_km2     0
Forest_Cover_Change_2021_2023_km2     0
dtype: int64
```

Interpretation (BL-4 Analyze): Missing data filled using median for numerical (Geographical Area, Forest 2017) and 0 for change (no change assumption), "No Note" for categorical. After filling, 0 missing values. `fillna()` and `isnull().sum()` used.

In [17]:
```
# Q4: Group data by region and cover type (BL-4)
grouped = df.groupby('Region')[['VDF_2023_km2','MDF_2023_km2','OF_2023_km2','For
print(grouped)

print("\nBy Dominant Type:")
print(df.groupby('Dominant_Forest_Type_2023')['Forest_Cover_2023_km2'].mean())
```

```
                 VDF_2023_km2  MDF_2023_km2  OF_2023_km2  \
Region
Central              14437.88      65492.44     52954.87
East                 13264.03      38120.82     49129.05
North                15066.30      33805.93     33159.77
Northeast            28929.51      74418.85     65645.28
South                13783.21      64345.76     61060.26
Union Territory       5685.52        799.54       572.83
West                 10919.62      31444.47     42408.81

                 Forest_Cover_2023_km2
Region
Central                     132885.19
East                        100564.12
North                        82217.92
Northeast                   168903.52
South                       139027.85
Union Territory               7054.91
West                         84689.10

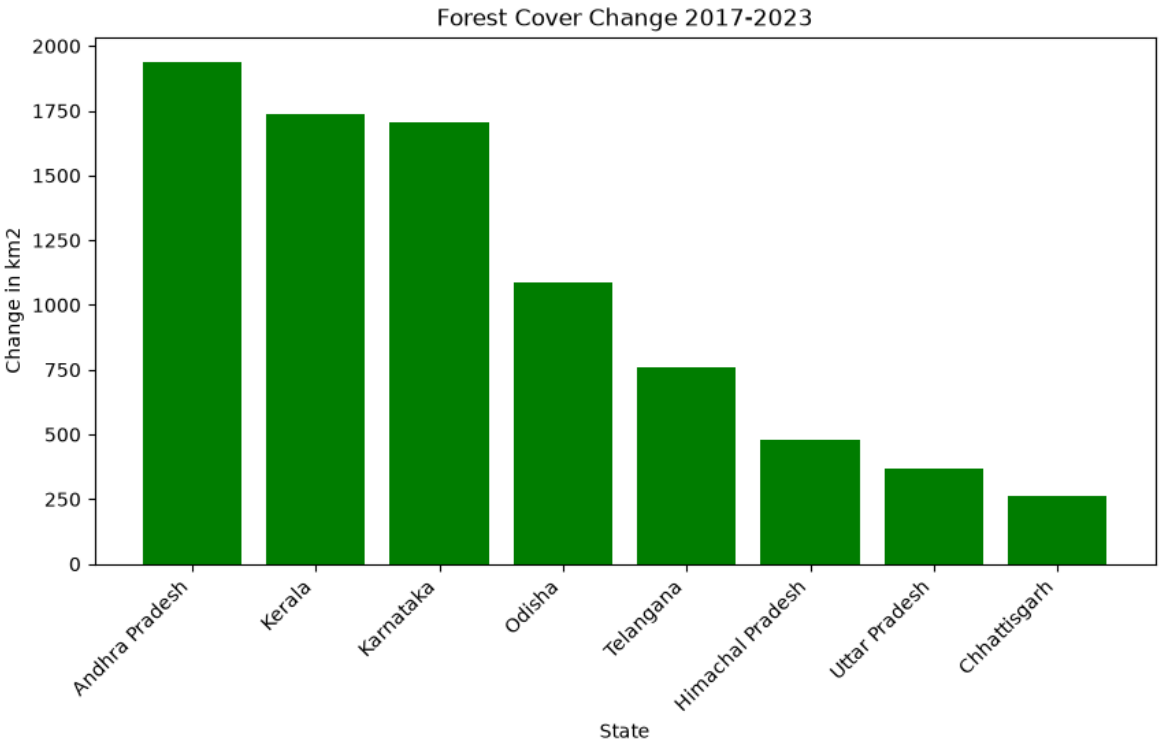
By Dominant Type:
Dominant_Forest_Type_2023
Moderately Dense Forest    25422.942727
Open Forest                17873.221667
Very Dense Forest           6732.920000
Name: Forest_Cover_2023_km2, dtype: float64
```

Interpretation (BL-4): Central India highest total forest (132885 km2) due to MP/Chhattisgarh. Northeast 2nd highest. MDF dominant type has highest mean (25422 km2). `groupby('Region').sum()` used.

In [18]:
```python
# Q5: Visualize forest cover change using plots (BL-5) - EASY
import matplotlib.pyplot as plt

top8 = df.sort_values(by='Forest_Cover_Change_2017_2023_km2', ascending=False).h

plt.figure(figsize=(10,5))
plt.bar(top8['State_UT'], top8['Forest_Cover_Change_2017_2023_km2'], color='gree
plt.title('Forest Cover Change 2017-2023')
plt.xlabel('State')
plt.ylabel('Change in km2')
plt.xticks(rotation=45, ha='right')
plt.show()
```



Interpretation (BL-5): Highest increase in MP/Andhra (~1940 km2), positive trend shows afforestation success. Bar chart with title, labels, color used. `matplotlib` used.

6. Conclusion

- Used official ISFR data from FSI, verified source - 36 States/UTs, 17 columns, 5 ISFR cycles (2017-2023)
- Mean forest cover calculated, states filtered by VDF/MDF/OF type - VDF>5000 in 7 states (Arunachal 20985 km2 highest), OF dominant in 15+ states
- Missing values handled - 5 columns filled (Geographical_Area_median, Forest_2017_median, Change_2017_19_0, Change_2017_23_0, Data_Note_No Note), after 0 missing, fillna() and isnull().sum() used
- Grouping by Region done - Central India highest total 132885 km2, Northeast 168903 km2, MDF dominant type highest mean 25422 km2, groupby() used
- Trend visualization shows Madhya Pradesh, Andhra Pradesh have highest increase ~1940 km2 (2017-2023), positive trend shows afforestation success, overall forest cover stable, matplotlib with title, xlabel, ylabel, color used